

6G PG Class-Lecture-13

Transcript

<https://otter.ai/u/dIEEoGqivXSYMW7QXhT5FR3FXYM?view=summary>

Overview

Prof. Neelkanth Kundu discussed the entanglement-assisted classical capacity, focusing on the generalization of super dense coding to arbitrary channels. He explained that Alice and Bob share entangled qubits, allowing Alice to transmit two bits of classical information per qubit use. The noiseless qubit channel's capacity was found to be two bits per channel use, doubling the Hollywood capacity of one bit. For the completely depolarizing channel, even with entanglement, the capacity remained zero. The bit flip channel's capacity was calculated as two minus $h_2(p)$, showing a one-bit gain with entanglement. The discussion also touched on the private capacity, which accounts for information accessible to eavesdroppers.

Outline

Entanglement-Assisted Classical Capacity Overview

- Prof. Neelkanth Kundu begins the meeting by discussing the previous class's topic on entanglement-assisted classical capacity.
- The main result of entanglement-assisted classical capacity is revisited, focusing on the generalization of the super dense coding protocol.
- Alice and Bob share a pre-shared entanglement, allowing Alice to convey two bits of classical information per qubit use.
- The generalization of super dense coding to arbitrary channels is introduced, termed entanglement capacity, where Alice and Bob share copies of pure entangled states.

Detailed Explanation of Entanglement Capacity

- Prof. Neelkanth Kundu explains the concept of entanglement capacity, where Alice encodes a message using an entangled qubit and transmits it over multiple uses of the channel.
- The maximum entanglement-assisted classical capacity is given by the maximum quantum mutual information.
- The quantum mutual information is calculated by replacing classical entropies with quantum entropies.
- The concept of purification is used to quantify how much information the channel preserves.

Discussion on Multiple Uses of the Channel

- Speaker 1 asks about the interpretation of multiple uses of the channel, questioning if it can be over time or different spatial domains.

- Prof. Neelkanth Kundu confirms that multiple uses can be over time slots, frequency domains, or spatial resources.
- The term "n usage of the channel" is clarified, indicating multiple uses of the channel to transmit one message.
- The representation of the signal sent over multiple channels is discussed, including encoding mechanisms and time slots.

Noiseless Qubit Channel Example

- The first example discussed is the noiseless qubit channel, where the output density operator remains unchanged.
- The noiseless channel is equivalent to super dense coding, resulting in a classical data transmission rate of two bits per channel use.
- The maximally entangled state, Bell state, is used to illustrate the noiseless channel example.
- The quantum mutual information is calculated, showing that the entanglement-assisted classical capacity is two bits per channel use.

Completely Depolarizing Qubit Channel Example

- The completely depolarizing qubit channel is introduced, where the channel replaces the input quantum state with the maximally mixed state.
- The Hollywood capacity of this channel is zero, as the channel completely removes all quantum information.
- The question is posed if entanglement can improve the capacity of this channel.
- The output density operator after the completely depolarizing channel is a product state, indicating that entanglement cannot improve the capacity.

Bit Flip Channel Example

- The bit flip channel is explained, where the channel applies the bit flip operator with probability p and preserves the quantum state with probability one minus p .
- The output density operator for the bit flip channel is calculated, showing that the state remains pure if no bit flip occurs.
- The quantum mutual information is calculated, showing that the entanglement-assisted classical capacity is two minus h_2 of p .
- The Hollywood capacity is compared, showing an additional one bit gain by using entanglement.

Private Capacity Introduction

- Prof. Neelkanth Kundu introduces the concept of private capacity, focusing on secure communication between Alice and Bob.
- The goal is to ensure high fidelity communication between Alice and Bob while minimizing information accessible to the eavesdropper.
- The private capacity is defined as the maximum mutual information between Alice and Bob minus the maximum information accessible to the eavesdropper.
- The setting involves Alice encoding a message and transmitting it over multiple uses of the quantum channel, with Bob decoding the message reliably.

Eavesdropper's Information Access

- The eavesdropper's information access is characterized by the eavesdropper channel, which maps Alice's quantum state to the combined state accessible to Bob and the eavesdropper.
- The main channel and the complementary channel are defined, with the main channel being the channel from Alice to Bob and the complementary channel being the channel from Alice to the eavesdropper.
- The partial trace is used to find the density operators for Bob and the eavesdropper.
- The private capacity is calculated by subtracting the maximum information accessible to the eavesdropper from the mutual information between Alice and Bob.

Comparison with Hollywood Capacity

- The private capacity is compared with the Hollywood capacity, showing that the private capacity is always less than the Hollywood capacity.
- The private capacity analysis is used in quantum key distribution (QKD) for secrecy analysis.
- The next module will focus on QKD protocols and applying classical and quantum information theory concepts to quantify the secret key rate.
- The importance of secure communication and minimizing information accessible to the eavesdropper is emphasized.

QKD Protocols and Secret Key Rate

- The next module will cover QKD protocols and the application of classical and quantum information theory to quantify the secret key rate.
- The concepts of cascade error correction leakage and privacy amplification leakage are introduced.
- The importance of post-processing in QKD to ensure security is discussed.
- The agreement between Alice and Bob on the final key length is highlighted.

Clarifications and Additional Questions

- Nakul Agrawal asks about the relationship between the current topic and previous semester's QKD lab.
- Prof. Neelkanth Kundu explains the concepts of cascade error correction leakage and privacy amplification leakage in the context of QKD post-processing.
- The importance of discarding bits to ensure security in QKD is discussed.
- The agreement between Alice and Bob on the final key length is reiterated, emphasizing the need for initial qubits to ensure a secure final key.